Leveraging Large Language Models and Feature Engineering for Robust Sentiment Classification of Informal Vietnamese Social Media

*Note: Sub-titles are not captured in Xplore and should not be used

1st Thi Mai Trang Nguyen
Posts and Telecommunications Institute of Technology
Hanoi, Vietnam
trangntm@ptit.edu.vn

2nd Vu Ngoc Son
Posts and Telecommunications Institute of Technology
Hanoi, Vietnam
sonvn.b22kh104@ptit.edu.vn

3rd Dinh Cong Nguyen
Posts and Telecommunications Institute of Technology
Hanoi, Vietnam
nguyendc.b22h082@ptit.edu.vn

4th Pham Van Minh
Posts and Telecommunications Institute of Technology
Hanoi, Vietnam
minhpv.b22kh078@ptit.edu.vn

5th Nguyen Trung Kien
Posts and Telecommunications Institute of Technology
Hanoi, Vietnam
kiennt.b22kh060@ptit.edu.vn

***Abstract*—This study addresses the challenges of sentiment analysis in informal Vietnamese social media text by developing a system to classify over 7,200 Facebook comments into positive, negative, and neutral classes. The input data, characterized by non-standard language and "teen code" abbreviations, required extensive preprocessing. A key contribution is the integration of Large Language Models (LLMs), specifically used for generating a normalization dictionary for abbreviations and ensuring accurate data labeling. The modeling pipeline also incorporated oversampling to effectively mitigate class imbalance in the dataset. We evaluated traditional feature engineering methods (BoW, TF-IDF) against modern, Vietnamese-specific Word Embeddings (PhoBERT, E5, VE), utilizing Naive Bayes and Support Vector Machine (SVM) classifiers. Experimental results demonstrated the superiority of semantic representation techniques. The optimized model, combining Vietnamese Embedding (VE) features with SVM after data balancing, achieved consistent high performance, reporting an Accuracy, Precision, Recall, and F1-score of 0.76. This system provides a robust and effective solution for tracking public discourse and managing content in complex Vietnamese online communities.**

***Index Terms*—Sentiment Analysis, Spam Detection, Social Media, Natural Language Processing, Anomaly Detection**

I. Introduction

In the era of pervasive social media, public figures, particularly global football icons such as Lionel Messi and Cristiano Ronaldo, attract massive attention from online communities. On platforms like Facebook, thousands of comments are generated daily to express admiration, criticism, or engage in heated debates. Automatically classifying these comments into positive, negative, and neutral categories holds significant value for understanding user behavior, analyzing public opinion, and supporting applications in community management, recommendation systems, and media research.

Despite its potential, sentiment analysis of Vietnamese social media text presents unique challenges. User-generated comments are often short, informal, and inconsistent, characterized by non-standard spellings, abbreviations, “teen code,” emojis, and other non-linguistic symbols. These factors complicate text normalization and pose difficulties for conventional natural language processing (NLP) pipelines.

This study addresses these challenges by developing a sentiment classification system tailored to Vietnamese Facebook comments. The proposed approach incorporates comprehensive preprocessing, including normalization of abbreviations and teen code with the assistance of Large Language Models (LLMs). Additionally, oversampling techniques are employed to mitigate class imbalance within the dataset. For feature extraction, both traditional methods—Bag of Words (BoW) and TF-IDF—and modern semantic representations—Vietnamese-specific embeddings such as PhoBERT, E5, and Vietnamese Embedding (VE)—are evaluated. Classification is performed using two established algorithms, Naive Bayes and Support Vector Machine (SVM).

Experimental results highlight the superiority of semantic embeddings over traditional feature-based approaches. In particular, the combination of VE features with SVM, after balancing the dataset, consistently achieved strong performance, with Accuracy, Precision, Recall, and F1-score all reaching

Identify applicable funding agency here. If none, delete this.

0.76. These findings demonstrate the effectiveness of the proposed system as a robust solution for monitoring public discourse and managing online communities in the Vietnamese context.

II. Related Work

Sentiment analysis for Vietnamese has evolved from traditional feature-based approaches to modern transformer-based models. Early studies typically relied on lexical and statistical features such as *Bag-of-Words* (BoW) and *TF–IDF*, combined with classifiers like Naive Bayes or SVM [1], [2]. While these methods established strong baselines, they were limited in capturing semantic and contextual information, especially in noisy social media text.

The emergence of pre-trained embeddings and transformer models significantly improved performance. In particular, PhoBERT [3] has been shown to outperform both multilingual and traditional models in Vietnamese sentiment analysis. Recent works have demonstrated the effectiveness of fine-tuning PhoBERT for sentiment classification tasks in domains such as product reviews and student feedback, often achieving state-of-the-art results [4], [5], [6]. These studies highlight the advantage of contextual embeddings over sparse feature engineering.

Another important research direction is handling the challenges of informal Vietnamese text. Social media data often contains "teen code," abbreviations, and emojis, which degrade model performance. Prior work emphasizes the need for normalization and preprocessing to address this issue [7]. Similarly, data imbalance is a persistent problem in multi-class sentiment analysis. Oversampling and data augmentation have been shown to improve classification metrics on Vietnamese datasets [8].

Despite these advances, relatively few studies directly compare traditional feature-engineering approaches against fine-tuned PhoBERT and other embeddings on highly informal Facebook comments. Moreover, the use of Large Language Models (LLMs) to automatically generate normalization dictionaries for "teen code" remains underexplored. Our work addresses these gaps by integrating LLM-assisted preprocessing with a systematic evaluation of both traditional and transformer-based approaches, including fine-tuned PhoBERT.

III. Methodology

The methodology is structured around robust data preparation applied to over 7,200 informal Vietnamese comments collected from Facebook. This process utilized LLMs to generate a normalization dictionary for "teen code" and perform automatic data labeling, along with oversampling to address class imbalance. We compared traditional feature methods (BoW, TF-IDF) against advanced Vietnamese-specific Word Embeddings (E5, VE, and PhoBERT). All feature vectors, including those from PhoBERT embeddings, were rigorously classified using classic machine learning algorithms: Naive Bayes and Support Vector Machine (SVM). In addition, we explored a fine-tuning approach with PhoBERT, which allows the model to adapt its deep contextual representations directly to the classification task, thereby capturing more nuanced semantic and syntactic patterns in Vietnamese text compared to fixed embeddings.

A. Data Collection and Processing Pipeline

The dataset for this study was constructed from more than 7,200 Vietnamese comments collected on Facebook posts related to Lionel Messi and Cristiano Ronaldo. Data acquisition was performed using the Apify platform, which provides automated web-scraping services for extracting comments from specific post links. The raw data were stored in `CSV` format to facilitate subsequent processing (Fig. **??**).

Named Entity Removal. Many users frequently tagged friends at the beginning of their comments, which introduced irrelevant noise into the dataset. To address this, we applied a Vietnamese Named Entity Recognition (NER) model, `NlpHUST/ner-vietnamese-electra-base`, to detect and remove personal names (PER entities). This step ensured that classification models focused solely on sentiment-bearing content rather than unrelated user references.

Teen-code and Abbreviation Normalization. Informal Vietnamese comments often contained non-standard spellings and abbreviations, such as "ko" → "không" or "dc" → "được." To normalize these forms, a Large Language Model (LLM), `gemini-2.0-flash-lite`, was used to automatically construct a dictionary mapping variants to standardized forms. The resulting dictionary included 792 unique mappings, which significantly reduced lexical noise and vocabulary sparsity.

Automatic Labeling with LLM. After normalization, sentiment labels were assigned using a more advanced LLM, `gemini-2.5-flash`. The model classified each comment into one of three categories: negative (-1), neutral (0), or positive (+1). To overcome token usage limits on Google's AI Studio, we employed 12 rotating API keys, with each request processing 100 comments. Labeling the full dataset of 7,200 comments required approximately 90 minutes. Manual inspection was performed afterward, and only about 100 labels required adjustment, typically due to sarcasm or context-specific nuances. This hybrid approach combined automation with human validation, yielding near-perfect labeling accuracy.

Preprocessing and Cleaning. Following annotation, a robust preprocessing pipeline was applied. The steps included lowercasing, removal of punctuation, emojis, numbers, and special characters, normalization of whitespace and elongated characters (e.g., "quáaaa" → "quá"), Vietnamese word segmentation, stopword removal, and elimination of incomplete records. These operations ensured consistent and analyzable input text.

Class Balancing. To address class imbalance, oversampling was applied by randomly duplicating minority-class instances until parity with the majority class was achieved. This step prevented classification bias and improved robustness across sentiment categories.

Dataset Splitting. Finally, the dataset was partitioned into training, validation, and test subsets using an 80:10:10 ratio. This division provided a reliable framework for model training, hyperparameter tuning, and unbiased evaluation of system performance.

B. Feature Engineering

To represent textual comments numerically, we employed both frequency-based and semantic-based feature extraction methods. These approaches provide the foundation for subsequent machine learning classification.

1) Bag of Words: The Bag-of-Words (BoW) model encodes each document as a sparse vector of word occurrence counts over the vocabulary. This representation is simple, computationally efficient, and effective with traditional classifiers such as Naive Bayes and SVM. However, BoW does not capture semantic information or word order, and its dimensionality grows rapidly with the vocabulary size.

2) TF-IDF: The Term Frequency–Inverse Document Frequency (TF-IDF) method extends BoW by weighting words according to their importance. Terms frequent in a specific document but rare across the corpus receive higher weights, improving discriminative power. TF-IDF typically yields better performance than raw BoW vectors, yet it still cannot capture context or semantic relationships between words.

3) Word Embedding: To overcome the limitations of sparse frequency-based methods, dense word and sentence embeddings were employed. Embeddings map tokens into continuous vector spaces where semantic similarity is preserved. In this study, three Vietnamese-specific embeddings were considered:

- **PhoBERT**: a transformer-based contextual model pre-trained on large-scale Vietnamese corpora.
- **E5**: a multilingual embedding model optimized for semantic similarity and retrieval tasks, producing sentence-level vectors.
- **Vietnamese Embedding (VE)**: a lightweight model specialized for Vietnamese, offering efficiency while maintaining strong performance in classification tasks.

Compared to BoW and TF-IDF, embedding-based methods capture richer syntactic and semantic information, making them more suitable for noisy, informal social media text.

4) TF-IDF weighted average word embeddings:

C. Classification Algorithms

For sentiment classification, we employed both probabilistic and margin-based learning algorithms. These models have been widely used in text classification tasks and serve as strong baselines when combined with frequency-based and embedding-based feature representations.

1) Naive Bayes: Naive Bayes (NB) is a family of probabilistic classifiers based on Bayes' theorem with the simplifying assumption of conditional independence among features. Despite this assumption rarely holding in practice, NB has proven effective in text classification due to its simplicity and robustness.

In this study, we adopted the Multinomial Naive Bayes variant, which is particularly suitable for text data represented as word counts or term frequencies. The model estimates class-conditional probabilities using maximum likelihood with Laplace smoothing:

$$P(y \mid x_1, \ldots, x_n) \propto P(y) \prod_{i=1}^{n} P(x_i \mid y), \tag{1}$$

$$\theta_{c,i} = \frac{\text{count}(w_i, c) + 1}{N_c + V}, \tag{2}$$

where $\text{count}(w_i, c)$ is the frequency of word w_i in class c, N_c is the total word count in class c, and V is the vocabulary size. The Multinomial NB is computationally efficient and performs well on high-dimensional sparse data such as BoW and TF-IDF vectors.

2) Support Vector Machine: Support Vector Machine (SVM) is a discriminative classifier that seeks an optimal hyperplane to separate data points of different classes while maximizing the margin between support vectors. Its effectiveness in high-dimensional spaces makes it particularly suitable for text classification.

Given feature vectors $\mathbf{x}_i$ with class labels $y_i \in \{-1, 1\}$, the linear SVM solves:

$$\min_{\mathbf{w},b} \ \frac{1}{2}\|\mathbf{w}\|^2 + C \sum_{i=1}^{n} \xi_i, \tag{3}$$

subject to $y_i(\mathbf{w} \cdot \mathbf{x}_i + b) \geq 1 - \xi_i, \ \xi_i \geq 0$, where C is a regularization parameter controlling the trade-off between maximizing the margin and minimizing classification errors.

We experimented with linear kernels, which are efficient for high-dimensional sparse representations, and also considered non-linear kernels (e.g., RBF) to capture more complex decision boundaries. SVM often outperforms Naive Bayes when combined with semantic embeddings such as PhoBERT, E5, and VE, thanks to its ability to exploit dense and context-rich feature spaces.

3) PhoBERT Fine-Tuning for Context-Aware Classification: To transcend the limitations of static, context-agnostic feature representations, we employed an end-to-end fine-tuning methodology centered on PhoBERT, a state-of-the-art transformer-based model pre-trained on a massive Vietnamese corpus. This approach operationalizes the principle of transfer learning, where generalized linguistic knowledge acquired from large-scale, unlabeled data is adapted to a specialized, downstream task—in our case, sentiment classification. Instead of treating the model's outputs as fixed feature vectors, fine-tuning allows the entire architecture, including its deep, multi-layered attention mechanisms, to be recalibrated for the specific semantic and syntactic nuances of informal Vietnamese social media text.

The fine-tuned model is conceptualized as a composite function, $F(X; \theta)$, where X represents an input sequence of tokens and θ denotes the set of all trainable parameters. This function is composed of two primary components: a

pre-trained encoder backbone, $\Phi(\cdot;\theta_{\text{base}})$, and a task-specific classification head, $g(\cdot;\theta_{\text{head}})$. The PhoBERT model serves as the encoder, which maps the input sequence X to a sequence of high-dimensional contextualized embeddings, $H = \Phi(X)$. From this sequence, an aggregate representation for the entire text, denoted as h_{cls}, is extracted—typically the embedding corresponding to the special `[CLS]` token. This vector is designed to encapsulate the holistic semantic content of the input.

$$h_{\text{cls}} = \Phi(X;\theta_{\text{base}})_{[\text{CLS}]} \tag{4}$$

This aggregate representation is then fed into the classification head, $g(h_{\text{cls}})$, which is typically a multi-layer perceptron. The function of this head is to project the rich, high-dimensional semantic space of PhoBERT onto the lower-dimensional output space of the task, yielding a vector of raw, unnormalized scores (logits), z, for each sentiment class.

The model's learning objective is to minimize the discrepancy between its predicted probability distribution and the ground-truth distribution of the labels. This is formally achieved by minimizing the Categorical Cross-Entropy loss function, $\mathcal{L}$, over the training dataset. For a single data instance (X, y), where y is the one-hot encoded true label vector, the loss is defined as:

$$\mathcal{L}(\theta) = -\sum_{c=1}^{C} y_c \log(p_c) \tag{5}$$

where C is the number of classes, and p_c is the predicted probability for class c. These probabilities are derived from the logits, z, via the Softmax function:

$$p_c = \text{softmax}(z)_c = \frac{\exp(z_c)}{\sum_{j=1}^{C}\exp(z_j)} \tag{6}$$

The full set of parameters, $\theta = \{\theta_{\text{base}}, \theta_{\text{head}}\}$, is optimized jointly using stochastic gradient-based methods, such as AdamW. The core of the fine-tuning process lies in the backpropagation of the loss gradient, $\nabla_\theta \mathcal{L}$, through both the classification head and the deep layers of the PhoBERT encoder. This holistic update mechanism allows the model's fundamental language understanding capabilities—its attention patterns and token representations—to become specialized for identifying sentiment-bearing linguistic cues. By doing so, the model learns to effectively interpret the complex and often ambiguous nature of informal discourse, including sarcasm, slang, and context-dependent expressions, thereby achieving a more robust and nuanced classification performance than is possible with static feature extraction methods.

IV. Experiments and Results

A. Evaluation Methodology

In order to rigorously evaluate the effectiveness of our proposed sentiment classification framework, we adopt a set of widely used evaluation metrics in natural language processing and text classification tasks. Given that our dataset is inherently imbalanced due to the distribution of sentiments in informal Vietnamese social media, relying solely on overall accuracy would be insufficient. Therefore, we employ additional metrics such as precision, recall, and F1-score, along with the confusion matrix for more comprehensive insights.

Confusion Matrix. The confusion matrix provides a detailed view of classification outcomes by reporting the number of true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN). This allows us to analyze how well the model differentiates between sentiment classes (negative, neutral, positive), as well as identify misclassification patterns.

Accuracy. Accuracy measures the ratio of correctly classified samples to the total number of samples. While it offers a general indication of performance, it can be misleading in imbalanced datasets since a classifier biased toward the majority class may still achieve high accuracy.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{7}$$

Precision. Precision evaluates the proportion of correctly predicted positive samples among all samples predicted as positive. High precision indicates that the model generates fewer false positives, which is crucial when the cost of misclassification is high.

$$Precision = \frac{TP}{TP + FP} \tag{8}$$

Recall. Recall, also known as sensitivity, measures the proportion of correctly predicted positive samples among all actual positive samples. High recall indicates that the model can capture most relevant instances, reducing false negatives.

$$Recall = \frac{TP}{TP + FN} \tag{9}$$

F1-score. The F1-score is the harmonic mean of precision and recall, providing a balanced evaluation metric that accounts for both false positives and false negatives. It is especially useful when dealing with class imbalance, as it prevents a misleadingly high score driven by only one of the two metrics.

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{10}$$

By combining these metrics, we obtain both a holistic and fine-grained understanding of model performance. In particular, accuracy provides a general overview, while precision, recall, and F1-score reveal the trade-offs between false positives and false negatives. The confusion matrix further allows us to interpret classification errors across sentiment classes, making the evaluation robust and reliable for the noisy, informal nature of Vietnamese social media data.

B. Experimental Setup

In this study, we conducted experiments on a dataset consisting of approximately 7,200 Vietnamese comments collected from Facebook. This dataset is characterized by a high degree of linguistic and semantic complexity, far exceeding that of typical text corpora. The comments are deeply embedded in the context of football, often containing implicit references,

inside jokes, and domain-specific jargon that require specialized knowledge to decipher. Furthermore, the language is highly informal, replete with contemporary slang, ephemeral neologisms (often referred to as 'teen code'), and figurative expressions such as wordplay and irony. Consequently, the surface-level text frequently masks the true underlying sentiment, demanding a high degree of inferential reasoning for accurate interpretation—many comments are cryptic and may be incomprehensible to an outside observer.

To manage this complexity during data preparation, we employed a two-agent approach leveraging Large Language Models. First, a lightweight agent powered by **Gemini 2.0 Flash** was utilized to systematically normalize abbreviations and 'teen code', reducing lexical sparsity. Subsequently, a more advanced agent based on **Gemini 2.5 Flash** performed the sentiment annotation. This model's sophisticated reasoning capabilities were crucial for accurately labeling the nuanced and context-dependent comments into three final categories: *toxic*, *neutral*, and *non-toxic*. Despite this careful preparation, the resulting dataset exhibited a significant class imbalance, a common characteristic of real-world social media data, which poses further challenges for model training.

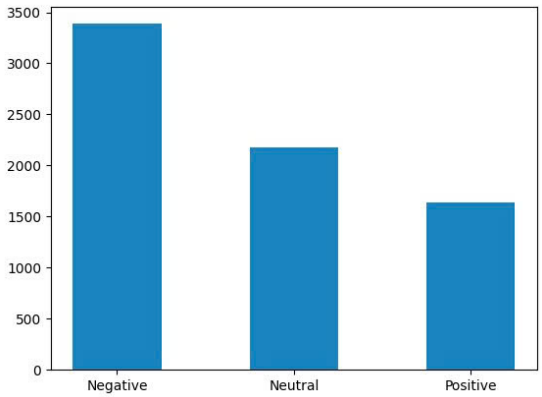

Figure 1: Class distribution

To address this issue, we compared model performance under two different scenarios: before applying oversampling and after applying oversampling. Oversampling was performed on the training set using the RandomOverSampler technique to balance the number of samples across classes. The validation and test sets were kept unchanged to preserve the natural distribution of the data and ensure fair evaluation.

For feature representation, we employed a variety of approaches, including:

- Bag-of-Words (BoW)
- Term Frequency - Inverse Document Frequency (TF-IDF)
- Word Embeddings: E5, Vietnamese-document-embedding (VE), PhoBERT

These feature extraction methods were combined with two widely used classification algorithms: Naive Bayes and Support Vector Machine (SVM). The dataset was split into training (70%), validation (10%), and test (20%) subsets. Models were trained on the training set, hyperparameters were tuned on the validation set, and final performance was evaluated on the test set.

For evaluation, we adopted standard metrics for multi-class classification, including Accuracy, Precision, Recall, and F1-score. In addition, Confusion Matrices were reported to provide deeper insights into the classification behavior across different classes. The experimental results are summarized in tables to facilitate a clear comparison of the effectiveness of each feature extraction method and classification algorithm, both before and after oversampling.

FE	Algorithm	Accuracy	Precision	Recall	F1
BoW	Naive Bayes	0.62	0.65	0.56	0.56
BoW	SVM	0.64	0.64	0.62	0.62
TF-IDF	Naive Bayes	0.51	0.57	0.38	0.31
TF-IDF	SVM	0.51	0.57	0.38	0.31
E5	SVM	0.67	0.68	**0.64**	**0.65**
VE	SVM	**0.68**	**0.69**	**0.64**	**0.65**
PhoBERT	SVM	0.67	**0.69**	0.63	0.64

Table 1: Results before Oversampling

FE	Algorithm	Accuracy	Precision	Recall	F1
BoW	Naive Bayes	0.69	0.69	0.69	0.69
BoW	SVM	0.75	0.76	0.75	0.75
TF-IDF	Naive Bayes	0.65	0.66	0.65	0.65
TF-IDF	SVM	0.69	0.70	0.69	0.69
E5	SVM	0.72	0.72	0.72	0.72
VE	SVM	**0.76**	**0.76**	**0.76**	**0.76**
PhoBERT	SVM	0.72	0.72	0.72	0.72

Table 2: Results after Oversampling

C. Fine-tuning PhoBERT for Sentiment Classification

In addition to using pre-trained embeddings as static features, we explored an end-to-end fine-tuning approach with the PhoBERT model. This method allows the model to update its internal weights and adapt its deep contextual representations specifically for our sentiment classification task. The model, whose architecture is detailed in the Methodology section, was evaluated on both the original imbalanced dataset and the oversampled training set to assess the impact of data balancing on its performance.

The results, summarized in Table **??**, show that the fine-tuned model achieved competitive performance. Before oversampling, the model reached a weighted F1-score of 0.70. After applying oversampling to the training data, performance saw a slight improvement, with the weighted F1-score increasing to 0.71.

Metric	Before Oversampling	After Oversampling
Accuracy	0.70	0.71
Macro F1-score	0.68	0.69
Weighted F1-score	0.70	0.71

Table 3: Performance of Fine-tuned PhoBERT Model

Interestingly, unlike the significant performance gains observed with traditional machine learning models (e.g., SVM

with BoW improved from 0.62 to 0.75), the impact of oversampling on the fine-tuned PhoBERT was marginal. This suggests that large transformer models may be inherently more robust to moderate class imbalance. A detailed analysis of per-class scores revealed that the model performed exceptionally well on the majority class (class 0, F1-score of 0.78 after oversampling) but was less effective in identifying the minority classes, particularly class 1 (F1-score of 0.61).

While the fine-tuned PhoBERT demonstrated strong capabilities, its overall performance did not surpass the best-performing traditional model. The combination of **VE + SVM after oversampling**, with an F1-score of **0.76**, remained the most effective solution in our experiments. This highlights that for certain datasets and tasks, a well-crafted feature engineering pipeline with a classic, robust classifier like SVM can outperform more complex deep learning models, offering a more computationally efficient yet powerful alternative.

V. Conclusion

Acknowledgment

References

[1] T.-T. Nguyen-Thi and A.-C. Duong, "A lexicon-based approach for sentiment analysis in vietnamese," in *Proceedings of ICCASA*, Springer, 2019.

[2] D. Q. Nguyen, A. T. Nguyen, and T. A. Vu, "Deep learning versus traditional classifiers on vietnamese students' feedback corpus," *arXiv preprint arXiv:1911.07223*, 2019.

[3] D. Q. Nguyen and A. T. Nguyen, "Phobert: Pre-trained language models for vietnamese," in *Findings of the Association for Computational Linguistics: EMNLP 2020*, 2020, pp. 1037–1042.

[4] V. T. Dang, A. T. Nguyen, and T. Pham, "Vietnamese sentiment analysis: An overview and comparative study of fine-tuning pretrained language models," *ACM Transactions on Asian and Low-Resource Language Information Processing*, 2023.

[5] M. Ngo, H. Ngo, and I. Valerievich, "Fine-tuned phobert for sentiment analysis of vietnamese phone reviews," *Can Tho University Journal of Science*, 2024.

[6] V. A. Nguyen, Q. Le, and M. Tran, "A hybrid phobert–cnn–lstm model for sentiment analysis of vietnamese student feedback," *Asian Journal of Research in Computer Science*, 2025.

[7] H.-Q. Nguyen, D. Q. Nguyen, and T. A. Vu, "Exploiting vietnamese social media characteristics for textual emotion recognition," *arXiv preprint arXiv:2009.11005*, 2020.

[8] K. Luu, M. Nguyen, and D. Nguyen, "An empirical study of text augmentation on social media text in vietnamese," *arXiv preprint arXiv:2009.12319*, 2020.